



SVKM's NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING /
SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING

Academic Year: 2021-22

Programme: B.Tech (Data Science)

Year: II Semester: IV

Subject: Statistical Structures in Data and Inference

Date: 09 April 2022

Marks: 100

Time: 10.00 am to 1.00 pm

Durations: 3 (hrs)

No. of Pages: 3

Final Examination

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	marks
CO1; SO2; BL1	a.	What is point estimation? Define the following with an example: 1. Estimator 2. Estimate	5
CO3; SO1; BL2	b.	Define steps performed in MLE with proper statistical justification.	5
CO1; SO1; BL1	c.	Mention advantage and disadvantages of non-parametric test.	5
CO2; SO6; BL4	d.	Define Adjusted R^2	5
Q2 CO3; SO1; BL5	a.	Define what is unbiasedness. Let X_1, X_2, X_3 be a random sample of size 3 drawn from the population with mean Θ test whether the following estimators are unbiased: 1. $T_1 = (X_1 + X_2)/2 + X_3$ 2. $T_2 = (X_1 + X_2 + X_3)/3$	10

		3. $T3 = (X1 + 3X2)/9 + 5X3$ 4. $T4 = (X1 - X2 + X3)$	
Q2 CO1; SO6; BL3	b.	1. Explain in brief what is Efficiency. 2. Define Relative efficiency and mention all 3 conditions for it. 3. Explain consistency in detail also mention the properties of consistency.	10
Q3 CO3; SO2; BL2	a.	1. Find MLE of parameter θ for the pdf $f(x, \theta) = (1 - \theta) \theta^{(x-1)}$ where $x = 0, 1, 2, \dots, \infty$ 2. Mention Properties of MLE.	10
Q3 CO3; SO1; BL3	b.	1. Find MLE of parameter θ for the pdf $f(x, \theta) = \theta(1 - \theta)^{x-1}$ where $x = 0, 1, 2, \dots, \infty$ 2. A sample of 400 male students is found to have a mean height of 67.47 inches. Can it be reasonably regarded as a sample from a large population with mean height 67.39 inches and standard deviation 1.30 inches. Test for 5% level of significance.	10
Q4 CO1; SO6; BL6	a.	1. Explain all the assumptions of linear regression and mention model of linear regression with explaining all the variables. 2. Mention the properties of OLS estimators.	10
Q4 CO4; SO7; BL3	b.	1. Explain the Durbin Watson test: i. Define test statistic ii. Explain the rules for autocorrelation iii. Assumptions of the Durbin Watson test	10
Q5 CO1; SO6; BL2	a.	1. What is logistic distribution function? 2. What is its significance in logistic regression model? 3. Explain logit transformation of logistic regression model	10
Q5 CO4; SO7;	b.	1. Explain the difference between Point Estimation and Interval Estimation. 2. Explain Heteroscedasticity	10

BL3			
Q6 CO1; SO1; BL1	a.	For Wilcoxon signed rank test (for one sample) mention the: 1. Assumptions 2. Hypothesis to be tested 3. Test statistic 4. Decision Rule	10
Q6 CO1; SO1; BL2	b.	For Mann whitney test mention the: 1. Assumptions 2. Hypothesis to be tested 3. Test statistic 4. Decision Rule	10